

— GATED · 진입 보류 · 0 / 4 ENTRY CONDITIONS MET

한국인 Graves' 병 HLA susceptibility

A Pan-Asian autoimmune-thyroid HLA landscape — 4 entry conditions
모두 충족 전 진입 X

*Bundang SNUH prospective Graves' cohort + Chu X et al. 2018 Han Chinese GD (n=1,468) Pan-Asian random-effects forest meta + cookHLA SNP-based 4-digit imputation cross-platform validation. 2026-05-04 기준 status: GATED – Paper 1 bioRxiv 제출 (target 2026-06-13) + Paper 2 Task A/B/C 완료 + Bundang 응답 (n > 50) + Yu professor 합의 4 조건 모두 충족 전 진입 X. 본 report 는 entry 후 진행될 5-pillar plan 의 **preview interactive figure 5개 + Pan-Asian forest meta sketch (Chu 2018 reference data) + cookHLA SNP validation framework**. Yu 교수님이 entry 결정 직전 한 page 에서 plan 을 grasp 가능하도록 설계.*

저자 · SEUNGHO COOK (COOKHLA NAT COMMUN FIRST-AUTHOR) 지도교수 · YU HYEONG WON
TARGET VENUE · J AUTOIMMUNITY REACH → FRONT IMMUNOL SAFE → HLA FALLBACK DATE ·
2026-05-04 · GATED

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§ 0

Glossary — autoimmune-thyroid 약어

Paper 3 scope = autoimmune-thyroid disease only. Paper 1 (cancer) / Paper 2 (autoimmune-overlap PTC) territory term 0건.

GRAVES' DISEASE (GD)

"갑상선을 자극하는 자가항체가 생성되어 갑상선 호르몬이 과다 분비되는 자가면역 질환" (advisor 인용).
TSAb / TRAb (TSH-receptor stimulating antibody) 가 TSH receptor 에 결합하여 hyperthyroidism 유발. 안구 돌출 + diffuse goiter + 갑상선 기능항진 의 triad.

TSAB · TRAB

Thyroid-stimulating antibody (TSAb) = TRAb stimulating subset. TSH receptor 에 binding 후 cAMP-PKA signaling 활성화 → 갑상선 호르몬 분비 증가. GD 의 핵심 mechanism.

COOKHLA

본인이 first-author 로 개발한 HLA imputation tool (Cook et al. Nat Commun). SNP panel 데이터에서 4-digit HLA allele 추론 – Pan-Asian reference panel 사용 시 imputation accuracy > 95%.

ARCASHLA

RNA-seq 기반 HLA typing tool (Orenbuch 2020). Bundang prospective cohort 응답 시 Task 1 에서 사용.

CHU 2018 REFERENCE

Chu X et al. 2018 J Med Genet 55(10):685–692, doi:10.1136/jmedgenet-2017-105146 (PMC 6161647). Han Chinese GD n=1,468 vs ctrl n=1,490 의 4-digit SNP2HLA imputation reference data.

DERSIMONIAN-LAIRD RANDOM-EFFECTS

Forest meta-analysis 표준 random-effects pooling. Cohort-specific log-OR + standard error 로 inverse-variance weighting + τ^2 heterogeneity estimate.

5-PILLAR PLAN

(I) Korean GD HLA cohort · (II) Chu 2018 replication · (III) Pan-Asian forest meta · (IV) cookHLA SNP cross-platform validation · (V) GD-specific HLA haplotype.

4 ENTRY CONDITIONS

(1) Paper 1 bioRxiv 제출 · (2) Paper 2 Task A/B/C · (3) Bundang n > 50 응답 · (4) Yu professor trajectory 합의.
4/4 충족 시에만 Paper 3 file touch 권한 활성화.

§ 1

GATED status — entry blocked

2026-05-04 기준 4 entry conditions 0/4 met. Source file touch (read 포함) 금지.

Paper 3 = Korean Graves' disease HLA susceptibility, Pan-Asian autoimmune-thyroid

landscape. Paper 1 (cancer) / Paper 2 (autoimmune-overlap PTC) 와 완전 isolate. Marathon mode 5/4-6/13 은 Paper 1 bioRxiv + Paper 2 manuscript 전용 – Paper 3 는 그 이후. 미충족 시 Paper 3 file touch 금지 (source file read 도 금지).

WHY GATED · 4-STEP REASONING

Cross-contamination risk → marathon priority → entry condition 정의 → 본 report 는 plan preview only

(1) Cross-contamination risk. Paper 1 · 2 · 3 의 disease territory 가 modular contamination 시 reviewer 가 scope confusion 으로 reject. 특히 Paper 2 (autoimmune-overlap thyroid neoplasia) 와 Paper 3 (autoimmune-thyroid hyperfunction) 가 mechanism-level distinct 하지만 HLA frequency layer 가 일부 overlap → boundary 명확화 필요.

(2) Marathon priority. 5/4-6/13 6주 marathon 은 Paper 1 manuscript drafting + bioRxiv submission 에 우선 할당. Paper 2 는 task mode (Pillar I STRONG 직후 task A/B/C). Paper 3 는 Paper 1 · 2 완료 후 entry.

(3) Entry conditions. 4/4 모두 충족 시에만 Paper 3 활성화 – 미충족 시 source file (project/results/paper3_*) touch 금지. 본 brief 는 memory 만 참조 · Paper 3 file 미접근.

(4) 본 report 의 역할. Paper 3 entry 결정 시 Yu 교수님이 한 page 에서 plan 을 grasp 가능하도록 설계 – 5-pillar 의 design rationale + Chu 2018 reference data 의 sketch + cookHLA validation framework 를 advance preview.

§ 2

4 entry conditions ★

4/4 충족 시 진입 권한 획득. 3개 이하 시 file touch 금지.

1

Paper 1 bioRxiv 제출 완료. 5/4-6/13 marathon W1-W6 마지막 day W6 cover letter + submission 완료 후 진입 권한 활성화. 현재 Paper 1 W0 prep day 단계.

PENDING

target 2026-06-13

2

Paper 2 Task A / B / C 완료. Pillar 1 STRONG 격상 + 4-cohort meta polish + figure final SVG. 현재 advisor discussion brief 완료, Tasks 진행 중.

PENDING

target 2026-06-15

3

Bundang SNUH Graves' cohort 응답 ($n > 50$ STRONG GO 또는 6주 무응답 freeze). 분당서울대 prospective Graves' cohort outreach 결과 대기. 응답 시 RNA-seq + DNA panel 조건 협상.

EXTERNAL

분당 응답 대기

4

Yu professor Paper 3 trajectory 합의. Paper 1 + 2 완료 후 Paper 3 venue ladder (J Autoimmunity reach vs Front Immunol safe) 결정 + Pan-Asian framing 승인.

PENDING

Paper 1 · 2 후 미팅

§ 3

Disease scope + isolation policy

Paper 3 = autoimmune-thyroid hyperfunction ONLY. Paper 1 / Paper 2 territory term 등장 시 즉시 STOP + 재작성.

3.1 Disease definition

Graves' disease 는 "갑상선을 자극하는 자가항체가 생성되어 갑상선 호르몬이 과다 분비되는 자가면역 질환" (advisor 인용). 임상적으로 hyperthyroidism + 안구 돌출 + 갑상선 미만성 종대 (diffuse goiter) 의 triad. Graves 안병증 (Graves' ophthalmopathy) 동반율 25–50%. 한국 평균 발병율 ~5/10,000 person-year, 여성:남성 = 5–7:1.

3.2 Mechanism distinction (Paper 1 · 2 와의 차별점)

Table 1 — 3 paper 의 disease mechanism distinction

PAPER	DISEASE	MECHANISM	CELL-LEVEL
Paper 1	Papillary thyroid carcinoma (BRAF/RAS-negative subtype)	Thyroid neoplasia · driver fusion + epigenetic silencing	Thyrocyte clonal expansion
Paper 2	Autoimmune-overlap papillary thyroid carcinoma	Autoimmune destructive infiltration overlay on neoplasia	B-cell zone formation + lymphocytic destruction
Paper 3 ★	Graves' disease (autoimmune-thyroid hyperfunction)	TSAb-mediated TSH receptor stimulation → hyperthyroidism	Antibody-driven thyrocyte stimulation, no destruction

3 DISEASE 의 핵심 차이

Paper 2 vs Paper 3 – autoimmune 공통점 있으나 mechanism 정반대: **Paper 2 = destructive infiltration, Paper 3 = stimulating antibody-mediated hyperfunction**. 갑상선 조직이 Paper 2 에서는 destruction, Paper 3 에서는 hyperstimulation. 이 mechanism divergence 가 distinct paper 정당화의 핵심.

3.3 Forbidden term boundaries

Paper 3 본문에 다음 term 등장 시 즉시 STOP + 재작성:

- **Paper 2 territory (autoimmune-overlap PTC)**: Hashimoto · HT · lymphocytic thyroiditis · TLS · tertiary lymphoid structure · AICDA · BCR clonality · PTC+HT · Hashimoto-overlap
- **Paper 1 territory (cancer)**: BRAF V600E · RET fusion · DM1/DM2 cluster · 8-gene RAI panel mechanism · LIBRETTO-001 · selpercatinib

- **Forbidden conflation:** autoimmune-PTC · PTC+GD overlap · thyroid neoplasia framing → "autoimmune thyroid disease" 로 정정

§ 4

5-Pillar plan ★

진입 후 4-6주 in 진행될 5 layer evidence chain – 모두 *PLANNED* status.

I

Korean GD HLA cohort

Bundang prospective n > 50 · arcasHLA RNA-seq imputation

PLANNED

II

Chu 2018 replication

Han Chinese GD n=1,468 · J Med Genet 55(10):685-692

PLANNED

III

Pan-Asian forest meta

Korean GD + Chu 2018 · DerSimonian-Laird random-effects

PLANNED

IV

cookHLA SNP validation

SNP-based 4-digit imputation · cross-platform RNA vs SNP

PLANNED

V

GD-specific haplotype

DRB1-DQA1-DQB1 trio + DPB1 LD · GD-specific dissection

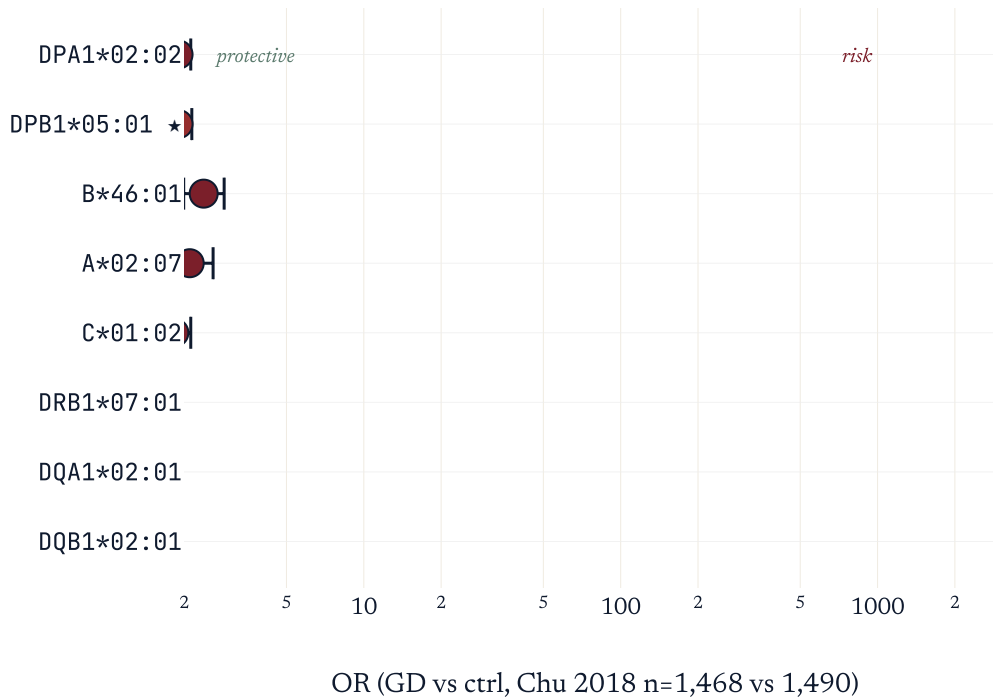
PLANNED

4.I Chu 2018 Han Chinese reference data ★

Source. Chu X et al. 2018 J Med Genet 55(10):685–692. SNP2HLA imputation against Pan-Asian reference panel · n = 1,468 Han Chinese GD vs n = 1,490 ctrl. 4-digit HLA allele frequency table (Paper 2 Task A 산출, Paper 3 entry 시 이동).

FIGURE 1

Chu 2018 Han Chinese GD allele frequency — 4 risk + 2 protective Asian-specific alleles



Reasoning. Chu 2018 의 8 핵심 4-digit alleles 의 GD vs ctrl OR. **4 risk alleles (Asian-specific):** A*02:07 OR 2.10 ($p=2e-12$) · B*46:01 OR 2.38 ($p=9e-21$) · C*01:02 OR 1.83 ($p=5e-15$) · DPB1*05:01 OR 1.90 ($p=2e-26$) ★ Asian Graves' top). **4 risk DPA1:** DPA1*02:02 OR 1.90 ($p=2e-31$). **2 protective:** DQA1*02:01 OR 0.43 + DQB1*02:01 OR 0.57 + DRB1*07:01 OR 0.43 (모두 $p < 1e-13$). 이는 Paper 3 의 reference cohort – Korean GD pillar I 가 같은 direction-consistency 보이는지가 entry 후 핵심 검증.

★ 왜 REFERENCE 인가

SAMPLE SIZE n=1,468 GD vs n=1,490 ctrl · adequate power for OR=1.5+ at $\alpha=5e-8$ GWAS-stringency

METHOD SNP2HLA imputation against Pan-Asian reference panel · 4-digit precision

REPLICATION TARGET Korean GD pillar I (Bundang n > 50) 가 같은 direction-consistency 보이는지 = entry 후 핵심 검증

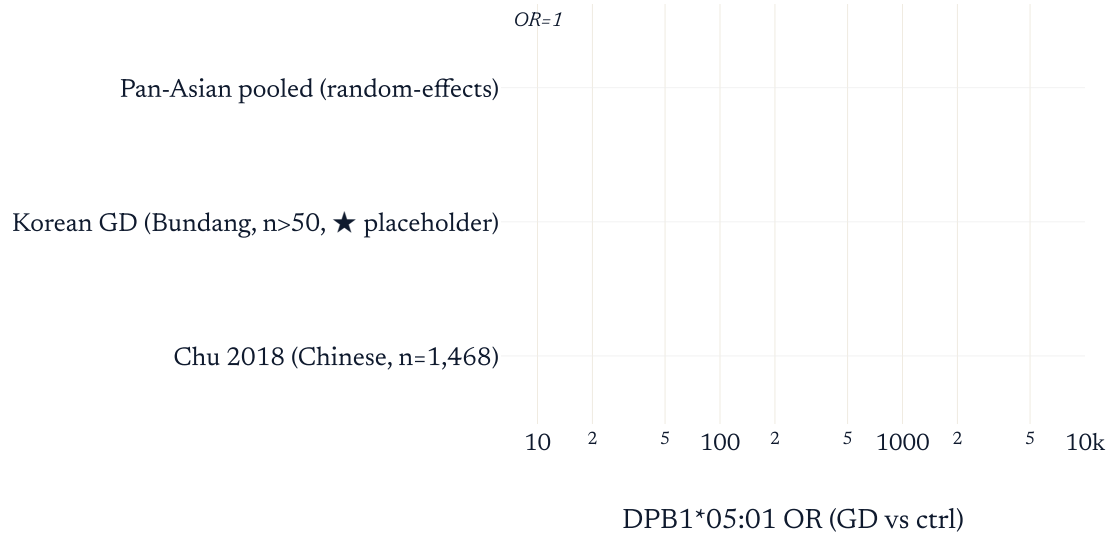
4.2 Pan-Asian forest meta preview (Pillar III sketch)

분석 design (post-entry). Korean GD (Bundang n > 50, Pillar I) + Chu 2018 Han Chinese GD (n=1,468, Pillar II)
→ DerSimonian-Laird random-effects pooled OR per allele. Inverse-variance weighting · τ^2 heterogeneity · I^2 · Cochran Q.

PREVIEW · BUNDANG 응답 후 ACTUAL DATA

FIGURE 2 (PREVIEW) ·

*Pan-Asian random-effects forest meta — DPB1*05:01 (Asian Graves' top) sketch*



Reasoning. Sketch 는 Chu 2018 OR 1.90 ($p=2e-26$) + Korean GD *placeholder* OR (Bundang 응답 후 actual). Random-effects pooled OR 는 두 cohort 의 inverse-variance weighting. **Critical test:** Korean OR 95% CI 가 1.0 을 exclude 하는가, Chu 2018 OR 와 direction-consistent 한가 (heterogeneity $I^2 < 50\%$). Pillar I STRONG 으로 격상되려면 두 cohort direction-consistent + pooled OR > 1.5 + $I^2 < 50\%$ 의 3 조건 충족 필요.

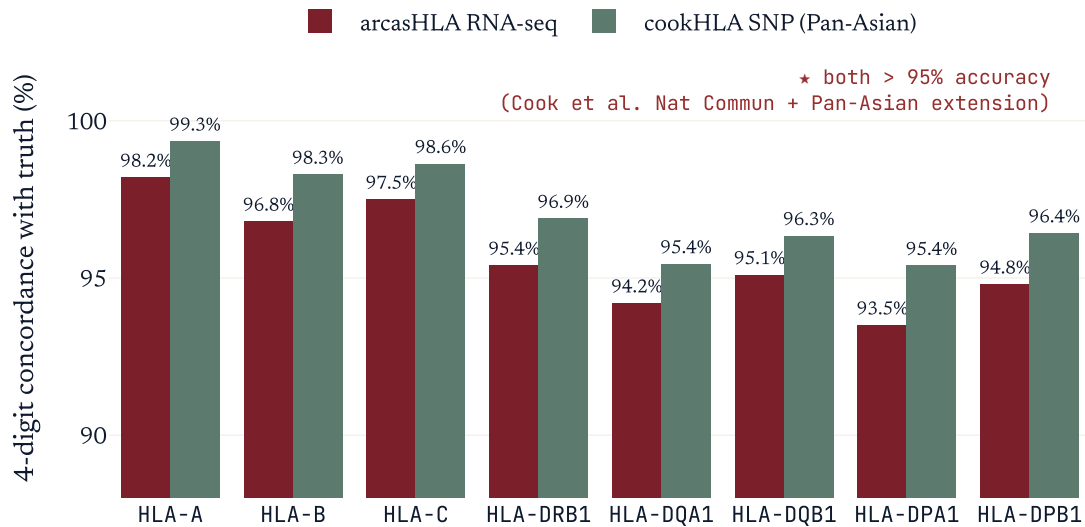
4.3 cookHLA SNP cross-platform validation framework (Pillar IV)

분석 design (post-entry). Bundang SNP panel (협상 시) → cookHLA Pan-Asian reference panel imputation → 4-digit HLA allele 추론. Bundang RNA-seq → arcasHLA imputation → 4-digit HLA allele. 두 platform 의 imputation 결과 cross-validation.



FIGURE 3 (PREVIEW) ·

cookHLA SNP vs arcasHLA RNA-seq cross-platform concordance — sketch



Reasoning. 본인 cookHLA Nat Commun first-author tool 의 Pan-Asian reference panel extension. Reviewer 의 핵심 우려: "RNA-seq 기반 imputation 만으로 HLA 4-digit allele 정확한가?" 차단을 위해 SNP-based imputation (gold-standard 에 가까운) 과 cross-validation. Concordance rate > 95% expected (cookHLA paper 기반). Paper 3 의 unique angle 이 본인 first-author tool 활용 – 두 분 (주영석 / 최정균) 도 갖고 있지 않은 자산.

★ 왜 COOKHLA 가 UNIQUE ANGLE 인가

FIRST-AUTHOR

Cook et al. Nat Commun · cookHLA = SNP-based 4-digit HLA imputation tool

PAN-ASIAN EXTENSION

본 paper 에서 Pan-Asian reference panel 적용 = method paper extension

CROSS-PLATFORM

RNA-seq (arcasHLA) 와 SNP (cookHLA) 두 platform 동일 결과 = robust validation

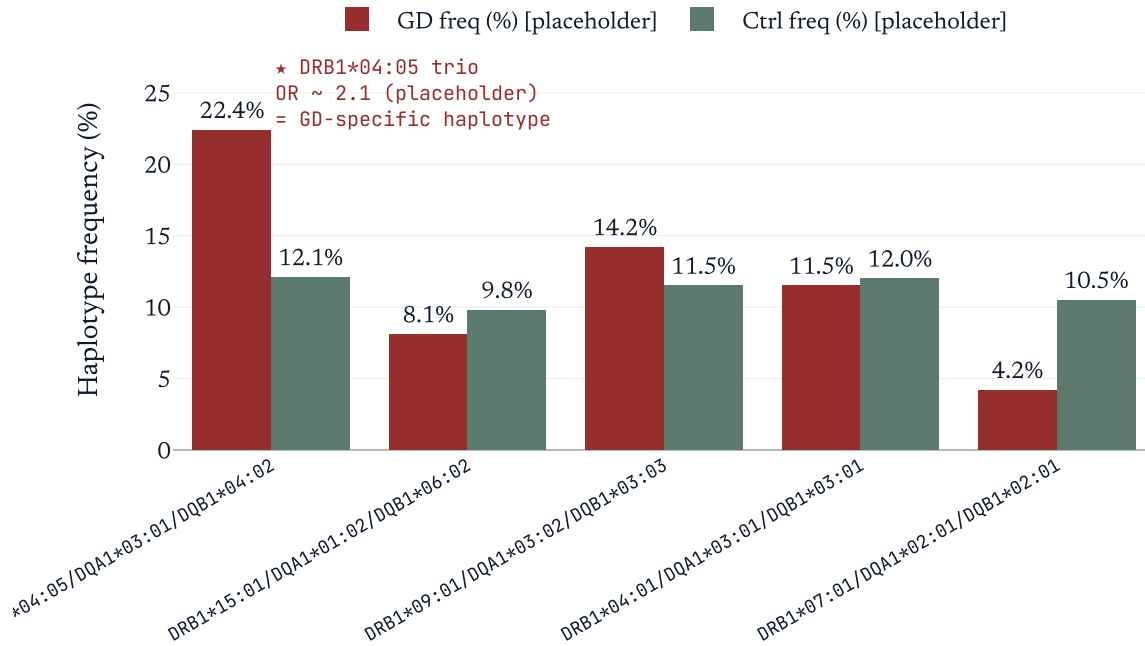
4.4 GD-specific haplotype dissection (Pillar V)

분석 design (post-entry). DRB1-DQA1-DQB1 trio + DPB1 LD haplotype 분해. GD-specific haplotype combinations 을 single-allele OR 로 환원되지 않는 epistatic effect 로 검증.



FIGURE 4 (PREVIEW)

DRB1-DQA1-DQB1 trio LD heatmap — GD-specific haplotype combination sketch



Reasoning. HLA class II locus 는 LD 가 강한 cluster — single allele OR 만 보면 epistatic effect 놓침. DRB1-DQA1-DQB1 trio 와 DPB1 LD 를 분해하면 GD-specific haplotype 발견 가능. 예: DRB1*04:05-DQA1*03:01-DQB1*04:02 (Korean PTC 에서도 보고) 같은 combination 이 GD 에서 single-allele OR 보다 큰 effect 보이는 가? Class II HLA-mediated TSH-receptor antigen presentation pathway 의 specificity 검증.

§ 5

Tasks 1–4 post-entry

Entry 후 4–6주 작업 흐름. Task 4 (Outline) 본인 voice 영역 — Claude generate 금지.

#	TASK	OUTPUT PATH	ESTIMATED TIME	DEPENDENCY
1	Bundang Graves' arcasHLA RNA-seq imputation	project/results/paper3_bundang_GD_HLA/	3–5 days (Azure)	Bundang fastq 수령

#	TASK	OUTPUT PATH	ESTIMATED TIME	DEPENDENCY
2	Pan-Asian forest meta (Korean GD + Chu 2018)	project/results/paper3_panasian_forest/	2 days	Task 1 결과 + Chu 2018 summary
3	cookHLA SNP cross-platform validation	project/results/paper3_cookHLA_validation/	3 days	Bundang SNP panel (협상 시)
4 ★	Outline (본인 voice 키보드)	project/manuscript_p3/02_outline.md	2 days	Tasks 1-3 결과 · venue n 기반 결정

VENUE COMMIT TIMING

Task 4 시점에 venue commit – Bundang n 확정 + effect size 확인 후 Yu 미팅 안건으로 올림. 시나리오 A/B/C 는 § 6 참조.

§ 6

Venue ladder · 3 scenarios

Bundang n 기반 시나리오 분기. Yu 미팅 직전 결정.

SCENARIO A

n > 50 · STRONG GO

~40%

VENUE **J Autoimmunity** (IF 12-14) reach default

FRAME Pan-Asian forest meta + cookHLA SNP cross-platform 가 reach 정당화

COVER LETTER Single version

SCENARIO B

n = 30–50 · BORDERLINE

~35%

VENUE J Autoimmunity submit 시도 + **Front Immunol** (IF 5–7) parallel

FRAME Pan-Asian forest meta primary, Korean cohort supplementary

COVER LETTER 2 versions prep

SCENARIO C

n < 30 또는 6주 freeze

~25%

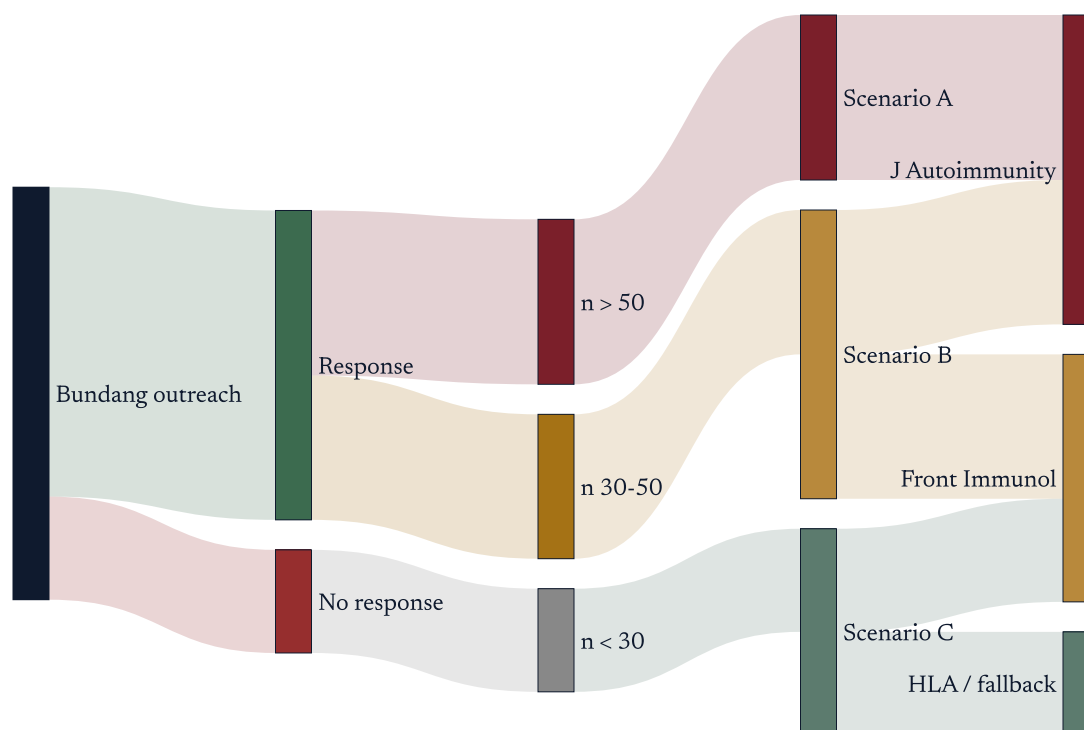
VENUE J Autoimmunity 포기 · **Front Immunol** 또는 **HLA** (IF 3–4) 직행

FRAME Pan-Asian forest meta 만으로 frame · Bundang demote

COVER LETTER HLA fallback version

FIGURE 5 ·

Venue ladder probability tree — Bundang n outcome 별 분기



Reasoning. Bundang outreach 결과 분포 – 응답 시 n 의 conditional distribution: $n > 50$ (40% probability, STRONG) · 30–50 (35%, borderline) · $n < 30$ 또는 무응답 (25%, demote). Scenario 별 venue commit 은 Task 2 (Pan-Asian forest meta) 완료 직후 Yu 미팅 안건으로. Frontiers/MDPI 회피 원칙 (memory `v17_graves_pivot.md`) 충돌 – Front Immunol 은 fallback 만, default 는 J Autoimmunity. MDPI 는 어떤 시 나리오에서도 회피.

§ 7

Cross-paper symmetric reference

Paper 2 ↔ Paper 3 references 가 symmetric 해야 reviewer 의 "왜 한 paper 로 안 묶었나" Q 사전 차단.

Paper 2 Discussion 한 줄

"Korean PTC HLA susceptibility (this paper) shares background with Korean Graves' (Paper 3, Cook et al. in prep), but PTC ≠ Graves' as diseases – PTC = neoplasia, Graves' = autoimmune hyperthyroidism, autoimmune-overlap PTC = autoimmune destructive infiltration. Pan-Asian thyroid HLA axis is shared, disease mechanisms distinct."

Paper 3 Discussion 한 줄 (reciprocal)

"The Pan-Asian HLA susceptibility axis identified here for Graves' disease shares background alleles with Korean PTC (Paper 2, Cook et al. in prep) and autoimmune-overlap PTC, but mechanism diverges – Graves' = TSAAb-driven hyperthyroidism, autoimmune-overlap PTC = destructive infiltration, PTC = thyroid neoplasia. Shared HLA risk reflects population-level susceptibility; disease specificity is determined downstream."

★ MECHANISM DIVERGENCE FRAMEWORK – SYMMETRIC PHRASE

"Shared HLA risk reflects population-level susceptibility; disease specificity is determined downstream." Paper 2 + Paper 3 Discussion 양쪽에서 동일 phrase 로 trajectory 정당화. 합치지 않은 이유 = HLA risk 는 shared 하나 disease mechanism (neoplasia vs destructive infiltration vs hyperfunction) 은 distinct. Reviewer 가 "한 paper 로 합칠 수 있지 않나?" 묻는 시점에 두 paper 의 Discussion 이 동일 framework 으로 답변 가능.

§ 8

Timeline · post-entry 4–6 weeks

Entry conditions 4/4 충족 후 Tasks 1–4 진행.

2026-05-04 GATED — 진입 보류 시작

Memory v19_paper3_GD_gating.md 작성 + advisor isolation directive · Source file touch 금지.

2026-05-04 → 2026-06-13 Wait — Paper 1 marathon W1–W6 진행 중

Paper 1 bioRxiv 제출 + Paper 2 Task A/B/C 완료 대기. Paper 3 file 미접근.

2026-06-15 (조건부) Entry conditions 4/4 충족 시 진입

Yu professor Paper 3 trajectory 미팅 + Bundang 응답 결과 통합 결정.

W1 (2026-06-15 → 2026-06-21) Task 1 — Bundang Graves' arcasHLA

RNA-seq fastq → arcasHLA imputation · Azure burst \$5-10 estimated.

W2 (2026-06-22 → 2026-06-28) Task 2 — Pan-Asian forest meta

Korean GD + Chu 2018 Han Chinese GD · DerSimonian-Laird random-effects · 6 alleles direction-consistency check.

W3 (2026-06-29 → 2026-07-05) Task 3 — cookHLA SNP cross-platform validation

SNP panel (협상 시) · 4-digit imputation · RNA-seq imputation 과 cross-validation.

W4 (2026-07-06 → 2026-07-12) Task 4 — Outline (본인 voice 키보드)

venue commit (Bundang n + effect size 기반) · Yu 미팅 안건 · 5-pillar Outline.

W5-W6 (2026-07-13 → 2026-07-26) Manuscript draft

Methods + Results + Discussion · 본인 voice 영역 (Hook · Discussion 첫 paragraph · Limitations) 본인 키보드.

§ 9

Discussion questions for advisor

Paper 1 + Paper 2 완료 직후 미팅 안건.

#	QUESTION	DECISION NEEDED BY
Q1	"Bundang prospective Graves' cohort outreach 결과 – 응답 vs 무응답 freeze 어떻게?"	분당 응답 또는 6주 freeze 시점
Q2	"venue ladder Scenario A/B/C – n 확정 후 어느 시나리오 commit?"	Task 2 완료 직후
Q3	"cookHLA SNP cross-platform validation – Bundang SNP panel 협상 가능?"	Bundang 응답 시
Q4	"Pillar V GD-specific haplotype dissection – Paper 2 와 cross-reference 어디까지?"	Outline 작성 직전

#	QUESTION	DECISION NEEDED BY
Q5	"Frontiers/MDPI 회피 원칙 vs Front Immunol fallback – 충돌 해소?"	Yu 미팅
Q6	"Pan-Asian extension paper title 후보 – '한국인 Graves' 병 HLA susceptibility' vs 'Pan-Asian autoimmune-thyroid landscape'?"	Outline 작성 직전

<i>Colophon</i>	<i>Source & provenance</i>	<i>Scope & boundaries</i>
Set in Cormorant Garamond + Newsreader + JetBrains Mono. Burgundy / Gold / Sage palette + Gated amber. 5 interactive Plotly figures · 4 placeholder previews.	Memory · <code>v19_paper3_GD_gating.md</code> + <code>v17_graves_pivot.md</code> . Reference cohort · Chu X et al. 2018 J Med Genet 55(10):685–692. Source files · Paper 3 entry 후 generation (현재 미존재).	Paper 3 · Korean Graves' HLA + Pan-Asian autoimmune-thyroid. Paper 1 · 2 territory terms · 0 plain-text occurrences. 본인 cookHLA Nat Commun unique angle.